

FORM-2
THE PATENTS ACT, 1970
(39 OF 1970)
AND
THE PATENT RULES, 2003
(As Amended)

COMPLETE SPECIFICATION
(SECTION 10; RULE 13)

TITLE
AI POWERED POTHOLE DETECTION

APPLICANT

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The following specification particularly describes the invention and the manner in which it is to be performed.

AI-POWERED POTHOLE DETECTION FOR INTELLIGENT ROAD INFRASTRUCTURE

Field of Invention

The aim of this invention is to explore The present invention relates to the field of road infrastructure monitoring and maintenance, and more specifically to an AI-powered pothole detection system. It uses computer vision and deep learning techniques, such as Convolutional Neural Networks (CNN) and YOLO models, to automatically identify potholes and road surface damages from images or video data. This invention aims to improve road safety and enable efficient, real-time detection of road defects.

Objectives of the Invention

The goal of this invention is to develop an AI-based system for automatic detection of potholes using image and video data. It aims to use deep learning techniques like CNN and YOLO for accurate and real-time detection, reduce manual inspection efforts, and improve road safety through early identification of road damages.

Background of the Invention

Maintenance of road infrastructure plays a vital role in ensuring safe and efficient transportation systems. Among various road defects, potholes are one of the most frequent and hazardous issues, primarily caused by factors such as heavy traffic loads, poor construction quality, aging infrastructure, and adverse weather conditions like rainfall and temperature variations. If not detected and repaired promptly, potholes can lead to serious road accidents, vehicle damage, increased fuel consumption, and higher maintenance costs for both governments and road users.

Conventionally, pothole detection has relied on manual inspection methods or complaints reported by the public. These approaches are not only time-consuming but also inefficient and unreliable, as they depend heavily on human effort and may result in delayed detection and response. In large urban areas with extensive road networks, continuous monitoring through manual methods becomes practically impossible.

KR20200071827A describes an artificial intelligence–based system for detecting potholes on road surfaces using visual data. The invention primarily relies on cameras or imaging devices to capture real-time images of the road, which are then analyzed using AI algorithms such as image processing and deep learning models. These algorithms are trained to recognize patterns associated with potholes and to distinguish them from other road features like cracks, shadows, or normal surface variations, thereby reducing false detections. Once a pothole is identified, the system can record and report its presence, potentially along with location information, for use in navigation systems or road maintenance planning. Compared to traditional methods that depend mainly on vehicle vibration or sensor data, this approach emphasizes “seeing” and intelligently interpreting road conditions, leading to more accurate and context-aware pothole detection

To overcome these limitations, several technological solutions have been proposed in recent years. For instance, systems described in patents such as **(US10235770B2)** focus on identifying potholes using vehicle-based sensors and image data. Similarly, **(US11221308B2)** introduces intelligent detection mechanisms aimed at improving the accuracy of pothole identification. Another approach, as seen in **(KR20200071827A)**, utilizes artificial intelligence and machine learning techniques to analyze road conditions and detect surface damages.

Although these existing solutions contribute significantly to the field, they still have certain limitations. Many systems depend on specialized hardware such as dedicated sensors, which increases implementation cost. Some approaches lack real-time processing capabilities, making them less effective for immediate action. Additionally, variations in lighting conditions, shadows, and road textures can reduce the accuracy of traditional image processing techniques.

With the rapid advancement of artificial intelligence, particularly deep learning models like Convolutional Neural Networks (CNN) and YOLO (You Only Look Once), there is an opportunity to develop more accurate and efficient pothole detection systems. These models are capable of processing large amounts of visual data and detecting objects in real time with high precision.

Therefore, there is a strong need for a robust, cost-effective, and real-time solution that can automatically detect potholes using widely available devices such as cameras. The present invention aims to address these challenges by leveraging advanced AI techniques to improve detection accuracy, reduce dependency on manual inspection, and enhance overall road safety and infrastructure management.

Summary of the Invention

The present invention relates to an advanced AI-powered system designed for the automatic detection of potholes and road surface damages to improve road safety and infrastructure management. The system employs deep learning techniques, particularly Convolutional Neural Networks (CNN) and YOLO (You Only Look Once) models, to analyze images and video streams captured through cameras installed on vehicles, mobile devices, or roadside units. By processing visual data in real time, the system is capable of accurately identifying potholes under varying lighting and environmental conditions.

In one aspect, the invention includes a data acquisition module that captures road images or video continuously, followed by a preprocessing module that enhances image quality and removes noise. The processed data is then fed into the trained AI model, which detects and classifies potholes based on learned patterns. Upon detection, the system generates alerts, marks the location of the pothole, and can optionally integrate with GPS modules to provide precise geographical information for maintenance authorities.

Another aspect of the invention is its ability to operate without the need for expensive or specialized hardware, making it a cost-effective solution compared to traditional sensor-based systems. The system can be easily integrated into existing smart city infrastructure and intelligent transportation systems for large-scale deployment. Additionally, it supports scalability, allowing continuous improvement through model training with new data.

Overall, the invention provides a reliable, efficient, and real-time solution for pothole detection, reducing dependence on manual inspection, minimizing road accidents, and enabling timely repair and maintenance of road infrastructure.

Detailed Description of the Invention

The present invention provides a comprehensive AI-powered framework for detecting potholes and road surface defects with high accuracy and efficiency. The system is designed to operate in real-time environments and can be deployed across multiple platforms such as vehicles, smartphones, and fixed roadside units. It combines computer vision, deep learning, and data communication technologies to create an intelligent and automated road monitoring solution.

In a preferred embodiment, the invention begins with a data acquisition unit that continuously captures images or video frames of road surfaces using cameras. These cameras may be integrated into moving vehicles (such as cars, buses, or maintenance vehicles) or fixed at strategic locations. The system supports multiple input formats and can handle high-resolution data streams. The captured data may include diverse environmental variations such as low light, shadows, rain, and occlusions caused by traffic.

The captured data is forwarded to a data preprocessing and enhancement module, where several image processing operations are performed. These include frame extraction from video streams, resizing to standard input dimensions, noise filtering, contrast enhancement, and normalization. In some implementations, data augmentation techniques such as rotation, flipping, and brightness adjustment may also be applied during training to improve model robustness.

The core component of the invention is the AI-based detection engine, which utilizes deep learning models such as Convolutional Neural Networks (CNN) and YOLO architectures. The model is trained using a large dataset of annotated images containing potholes and non-pothole road surfaces. During operation, the model processes each frame and performs object detection by identifying pothole regions, drawing bounding boxes, and assigning confidence scores. The system can also classify the severity of potholes based on size, depth estimation (if applicable), and area covered.

In an advanced embodiment, the invention includes a geo-tagging and mapping module that integrates Global Positioning System (GPS) data with detection results. Each detected pothole is associated with precise latitude and longitude coordinates. This information can be visualized on digital maps, enabling authorities to monitor pothole distribution and prioritize repairs based on location and severity.

The invention further comprises a data storage and cloud integration module, where detected information is stored in a centralized database. This allows historical analysis, trend identification, and performance evaluation. Cloud connectivity enables remote access, data sharing, and large-scale deployment across multiple regions. The system may also support integration with Geographic Information Systems (GIS) for enhanced visualization and decision-making.

Additionally, a notification and reporting module is included to generate real-time alerts. When a pothole is detected, the system can automatically notify concerned authorities through mobile applications, dashboards, or email notifications. Reports may include details such as image evidence,

location, timestamp, and severity level. This facilitates quicker response and efficient maintenance planning.

The system is designed to be scalable and adaptable, allowing it to function effectively in both urban and rural environments. It can be integrated with smart city infrastructure, traffic management systems, and autonomous vehicles. The model can be periodically retrained with new datasets to improve accuracy and adapt to changing road conditions.

The system may also include a communication module, which sends real-time alerts or notifications to relevant authorities or maintenance teams. These alerts can include details such as location, severity, and number of potholes detected. Additionally, the system can maintain a database of detected potholes for future analysis and reporting.

The invention supports real-time processing, enabling detection while the vehicle is in motion. This allows continuous monitoring of road infrastructure without interrupting traffic flow. The system is designed to function efficiently under various environmental conditions and can adapt to different road types and textures.

Furthermore, the invention emphasizes cost-effectiveness and ease of deployment, as it primarily relies on commonly available hardware such as cameras and smartphones instead of expensive specialized sensors. The system is optimized for performance, ensuring minimal computational requirements while maintaining high detection accuracy.

In summary, the present invention offers a robust, intelligent, and fully automated solution for pothole detection, combining real-time processing, accurate detection, geo-localization, and efficient reporting mechanisms. It significantly enhances road safety, reduces manual inspection efforts, and supports proactive maintenance of road infrastructure.

Brief description of Drawing

Figure 1, Data flow diagram

The provided diagram represents the workflow of the AI-POWERED POTHOLE DETECTION FOR INTELLIGENT ROAD INFRASTRUCTURE, showcasing the key components and their interactions.

Claim:

The scope of the invention is defined by the following claims:

1. A system/method for detecting potholes using artificial intelligence, said system/method comprising the steps of:

- a) The system starts by collecting road images and video data using cameras mounted on vehicles, mobile devices, or roadside units (1). The captured data is fed into the preprocessing module (2).
- b) The preprocessing module processes the input data by performing operations such as resizing, noise reduction, normalization, and contrast enhancement to improve image quality (3). The processed data is then provided to the trained deep learning model (4).
- c) The developed system is deployed on a cloud-based or edge computing platform, wherein the model is trained using labeled datasets of potholes and non-pothole images (5), and the dataset is divided into training and testing sets, preferably in a 70:30 ratio (6).
- d) The proposed invention analyzes input images and video frames using deep learning algorithms such as Convolutional Neural Networks (CNN) and YOLO for real-time pothole detection (7). The system identifies potholes by detecting patterns, textures, and shapes associated with road damage (8).

- 1. As mentioned in claim 1, the system features deep learning models such as CNN and YOLO trained to extract meaningful patterns and features from road images for accurate pothole detection.
- 2. According to claim 1, the system processes captured image and video data to identify potholes and generates feature representations, which are used to classify road surface conditions.
- 3. According to claim 1, based on the detection results, the system generates outputs including bounding boxes, confidence scores, and location details of detected potholes.
- 4. According to claim 1, the system adapts and improves over time using updated datasets and machine learning techniques to enhance detection accuracy under different environmental conditions.

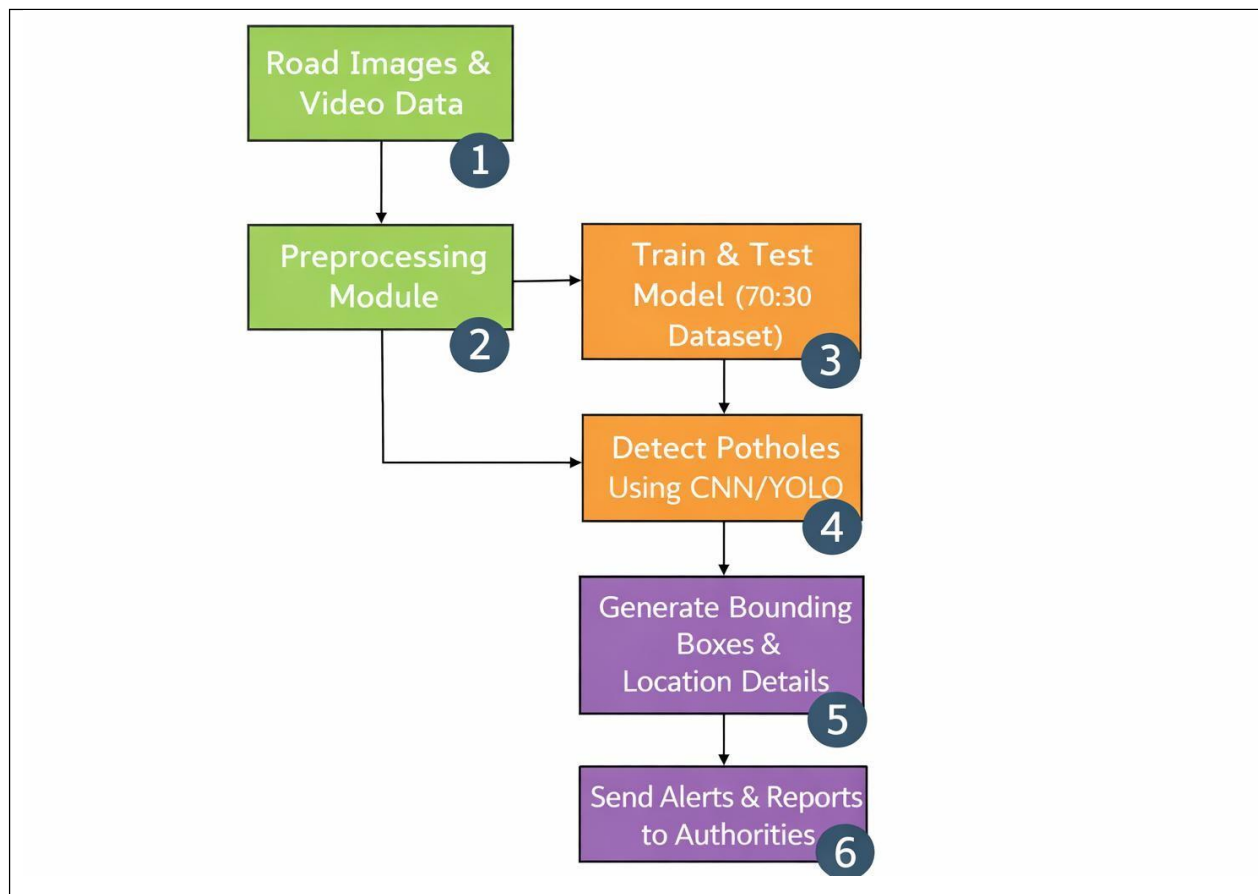


Figure 1 Data flow diagram

Abstract

Road infrastructure plays a crucial role in economic development and public safety. However, potholes and road surface damages are major challenges in urban and rural transportation systems. These defects often lead to traffic accidents, vehicle damage, increased fuel consumption, and traffic congestion. Traditional road inspection methods are manual, time consuming, costly, and inefficient. To address this issue, this project proposes an AI-based Pothole Detection System using Deep Learning techniques, specifically Convolutional Neural Networks (CNN) and YOLO (You Only Look Once) object detection models.